

PAPER_888: E(t) Full Lagrangian Unified Derivation

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Abstract

Full unified Lagrangian for the E(t) framework. $L_{\{E(t)\}} = E_{\text{net}}(t) \cdot V_{\text{filament}} \cdot S_{26}([SSq])$ combining both expansion and erosion branches with buoyancy sign-flipping and cosmological Λ link. The Lagrangian provides the complete variational principle for UQFF cosmological dynamics including GW constraints.

1. Core Equations

$$\begin{aligned} L_{\{E(t)\}} &= E_{\text{net}}(t) \cdot V \cdot S_{26} \\ E_{\text{net}} &= E_0 \cdot \exp(\kappa t + [SSq]t/26) \cdot S_{26} \cdot (2R-1) \\ \Lambda &= 8\pi G \cdot 0.692 \cdot \rho_{\text{crit}}/c^2 \end{aligned}$$

2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter
V_filament	1e48 m ³	Filament volume
F_UBi_over_FU	0.8	Buoyancy-to-field ratio

3. UQFF Integration

This calculator integrates `et_full_lagrangian.py` from Session 206 into the CP4 pipeline as class #472.

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